STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: MAC-04 | TAG: SHARP-LATHE-01

MAC-04: Sharp Manual Engine Lathe

Facility Zone: Zone 1: Manual Machining (MAC) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

CRITICAL ENTANGLEMENT WARNING: STRICTLY NO GLOVES WHILE SPINDLE / BLADE IS ROTATING

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. ANSI Z87.1 Safety Glasses
2. NO GLOVES (Rotating Chuck/Stock Entanglement Hazard)
3. No Rings, Watches, or Jewelry
4. Closed-Toe Shoes & Secured Hair/Clothing

2. PRIMARY OPERATIONAL HAZARDS

1. Severe entanglement hazard on rotating chuck, stock, and leadscrew
2. Flying chuck key projectile hazard if left in chuck
3. Cross-slide collision into rotating chuck jaws

3. BEFORE OPERATION CHECKLIST

1. Test emergency foot brake before starting spindle.
2. Verify 3-jaw/4-jaw/collet chuck clamping and REMOVE CHUCK KEY IMMEDIATELY from chuck before touching any controls.
3. Rotate chuck by hand to verify apron slide and compound toolpost clearance.

4. DURING OPERATION SAFETY RULES

1. Keep hands clear of rotating chuck and leadscrew at all times.
2. Use emergency foot brake for immediate stopping.
3. Never file, sand, or measure rotating workpieces without SRA supervision.

5. AFTER USE & SHOWROOM CLEANUP

1. Clean bedways and swarf pan with brush (never use compressed air).
2. Disengage power feed and half-nut levers.
3. Return all tooling, centers, and chuck keys to showroom condition.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Stomp emergency foot brake immediately and hit red E-Stop button; notify Lab Supervisor or SRA.